

Code-Layout, Readability and Reusability

Date	5 March 2026
Team ID	
Project Name	Smart Lender - code layout
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	yes	Proper indentation and formatting are maintained throughout the code.
2	Proper File Structure	Files and folders are logically organized	yes	Source code is organized into folders such as Frontend, Backend, ML Model, Database, and APIs.
3	Meaningful Variable Names	Variables reflect their purpose clearly	yes	Variables use descriptive names (e.g., loanAmount, fraudScore, applicantDetails).
4	Function / Method Names	Functions are descriptively named	yes	Functions have descriptive names such as predictLoanEligibility() and detectFraud().
5	Code Comments	Inline and block comments explain logic	yes	Important logic and machine learning workflow are

				documented with comments.
6	Modular Design	Code is split into reusable functions/modules	yes	Authentication, loan processing, fraud detection, and reporting are implemented as separate modules.
7	No Redundant Code	Duplicate or unused code is removed	yes	Common functions are reused, and duplicate code is avoided.
8	Error Handling	Exceptions and errors are handled gracefully	yes	Exceptions are handled using validation checks and error messages.

Reusable Components / Modules:

S.No	Component / Module Name	Language Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Authentication Module	Python	Login and Registration	High
2	Loan Processing Module	Python	Loan Application Workflow	High
3	Fraud Detection Module	Python, Scikit-learn	Loan Verification	High
4	Database Module	MySQL	Customer and Loan Records	High
5	Notification Module	Email API / Twilio	Approval, Rejection, OTP Alerts	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate modules.
Readability	5	Clear naming conventions, formatting, and comments.
Reusability	5	Modular components can be reused across different

		features.
Documentation / Comments	4	Important functions and workflows are documented.
Overall Score	4.75 / 5	The codebase is maintainable, scalable, and easy to extend.